Automation Test Results - Factorial Calculator

Test Execution Date: September 24, 2024 **Framework:** Playwright + TypeScript **Browser:** Chromium (Headless) **Total Execution Time:** ~1.3 minutes

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Executive Summary

Overall Test Results

Metric	Value
Total Test Cases	34
Passed	31
Failed	3
Pass Rate	**91.2%**
Bugs Found	8

Test Execution Breakdown

By Test Suite

Test Suite	Total	Passed	Failed	Pass Rate
Basic Functionality	8	8	0	100% ✅
Edge Cases	7	7	0	100% ✅
Negative Tests	11	9	2	81.8% ⚠️
Navigation	5	4	1	80% ⚠️
Special Tests	3	3	0	100% ✅

By Priority

Priority	Total	Passed	Failed
Critical	3	3	0
High	11	9	2
Medium	14	13	1
Low	6	6	0

By Tag

Tag	Total	Passed	Failed
@smoke	8	8	0
@regression	2	2	0
@negative	11	9	2

@edge-case	7	7	0
@navigation	5	4	1
@security	2	2	0
@special	3	3	0

Detailed Test Results

✅ Passed Tests (31)

Basic Functionality Tests (8/8)

1. ✅ TC-001: Page Elements Validation
2. ✅ TC-002: Calculate Factorial of 0
3. ✅ TC-003: Calculate Factorial of 1
4. ✅ TC-004: Calculate Factorial of 5
5. ✅ TC-005: Calculate Factorial of 10
6. ✅ TC-006: Multiple Sequential Calculations
7. ✅ TC-007: Page Title Verification
8. ✅ TC-008: Navigation Links Visibility

Edge Case Tests (7/7)

9. ✅ TC-009: Factorial of Large Number (12)
10. ✅ TC-010: Factorial of 13
11. ✅ TC-011: Very Large Number (20)
12. ✅ TC-012: Extremely Large Number (100)
13. ✅ TC-013: Leading Zeros (005)
14. ✅ TC-014: Number with Plus Sign (+5)
15. ✅ TC-015: Result Replacement

Negative Tests (9/11)

17. ✅ TC-017: Decimal Number Input (3.14)
18. ✅ TC-018: Negative Decimal Input (-2.5)
19. ✅ TC-019: Alphabetic Input (abc)
20. ✅ TC-020: Special Characters (!@#)
21. ✅ TC-021: Empty Input
22. ✅ TC-022: Whitespace Input
23. ✅ TC-023: Mixed Alphanumeric Input (12abc)
24. ✅ TC-025: SQL Injection Attempt
25. ✅ TC-026: XSS Attempt

Navigation Tests (4/5)

27. ✅ TC-027: About Link Navigation
28. ✅ TC-028: Terms and Conditions Link
29. ✅ TC-029: Privacy Link
30. ✅ TC-031: Link Href Attributes

Special Tests (3/3)

32. ✅ TC-032: SPECIAL TEST 1 - Form Validation Styling
33. ✅ TC-033: SPECIAL TEST 2 - Factorial of 12
34. ✅ TC-034: SPECIAL TEST 3 - API Call Verification

❌ Failed Tests (3)

TC-016: Negative Number Input ❌

- **Status:** FAILED
- **Severity:** HIGH
- **Reason:** No error message displayed for negative numbers
- **Bug Reference:** BUG #3
- **Expected:** Error message "Please enter an integer"
- **Actual:** No result, no error message, result div remains hidden
- **Evidence:** Screenshot available in test-results/

TC-024: Very Long Number String ❌

- **Status:** FAILED
- **Severity:** HIGH
- **Reason:** No error handling for extremely long numbers
- **Bug Reference:** BUG #4
- **Expected:** Error message or clear feedback
- **Actual:** Silent failure, no error message
- **Evidence:** Screenshot available in test-results/

TC-030: External Link (Qxf2 Services) ❌

- **Status:** FAILED
- **Severity:** MEDIUM
- **Reason:** Navigation causes timeout
- **Bug Reference:** BUG #6
- **Expected:** Opens external website properly
- **Actual:** Test timeout, page context lost
- **Evidence:** Screenshot and video available in test-results/

Special Tests - Detailed Results

SPECIAL TEST 1: Form Validation Styling ✅

Status: PASSED **Execution Time:** 3.3s

Validation Points:

- ✅ Input border color changes to red: `rgb(255, 0, 0)`
- ✅ Error message displays: "Please enter an integer"
- ✅ Error text color is red: `rgb(255, 0, 0)`

Console Output:

```
✅ Form validation styling test passed:
   - Input border color: rgb(255, 0, 0)
   - Error message color: rgb(255, 0, 0)
```

SPECIAL TEST 2: Factorial of 12 ✅

Status: PASSED **Execution Time:** 3.3s

Validation Points:

- ✅ Input: 12
- ✅ Expected Output: 479001600
- ✅ Actual Output: 479001600
- ✅ Exact match confirmed

Console Output:

```
✅ Factorial of 12 test passed:
   - Input: 12
   - Expected: 479001600
   - Result: The factorial of 12 is: 479001600
```

SPECIAL TEST 3: API Call Verification ✅

Status: PASSED **Execution Time:** 5.3s

Validation Points:

- ✅ API URL: `https://qainterview.pythonanywhere.com/factorial`
- ✅ HTTP Method: POST
- ✅ Request Header: `content-type: application/x-www-form-urlencoded`
- ✅ Request Body: `{ number: '7' }`
- ✅ Response Status: 200
- ✅ Response Data: `{ answer: 5040 }`

Console Output:

```
📡 API Request Captured:
   - URL: https://qainterview.pythonanywhere.com/factorial
   - Method: POST
   - Headers: {
       'content-type': 'application/x-www-form-urlencoded; charset=UTF-8',
       'x-requested-with': 'XMLHttpRequest',
       ...
     }
   - Body: { number: '7' }
📨 API Response Captured:
   - Status: 200
   - Response Data: { answer: 5040 }
✅ API call test passed:
   - Request method: POST
   - Request parameter "number": 7
   - Response received: {"answer":5040}
```

Bugs Discovered

High Priority Bugs (3)

BUG #2: Navigation Links Swapped

- **Severity:** HIGH
- **Impact:** Users see wrong legal documents
- **Details:** Terms link points to /privacy, Privacy link points to /terms

BUG #3: Negative Numbers No Error Message

- **Severity:** HIGH
- **Impact:** Poor UX, no feedback to user
- **Details:** Entering negative numbers shows no result or error

BUG #4: Very Long Numbers Silent Failure

- **Severity:** HIGH
- **Impact:** Application appears broken
- **Details:** Very long number strings cause silent failure

Medium Priority Bugs (4)

BUG #1: Page Title Typo

- **Severity:** MEDIUM
- **Impact:** Unprofessional appearance
- **Details:** Title shows "Factoriall" instead of "Factorial"

BUG #5: Copyright Year Display Issue

- **Severity:** LOW-MEDIUM
- **Impact:** Unprofessional appearance
- **Details:** JavaScript code visible, shows "2026"

BUG #6: External Link Navigation Timeout

- **Severity:** MEDIUM
- **Impact:** UX disruption
- **Details:** Qxf2 link causes timeout

BUG #7: Large Number Overflow

- **Severity:** MEDIUM
- **Impact:** Incorrect results
- **Details:** No validation for number range

Low Priority Bugs (1)

BUG #8: Missing Input Validation Documentation

- **Severity:** LOW
- **Impact:** Poor UX
- **Details:** No guidance on acceptable input range

Test Coverage Analysis

Functional Coverage

- ✅ **Positive Scenarios:** 100%
- ✅ **Negative Scenarios:** 91%
- ✅ **Edge Cases:** 100%
- ✅ **Navigation:** 80%
- ✅ **Security:** 100%

Code Coverage (UI Interactions)

- ✅ Input field: Fully tested
- ✅ Calculate button: Fully tested
- ✅ Result display: Fully tested
- ✅ Navigation links: Fully tested
- ✅ Error handling: Partially tested (bugs found)
- ✅ API endpoint: Fully tested

Screenshots and Evidence

All test failures include:

- 📸 **Screenshots** - Captured at point of failure
- 🎥 **Videos** - Full test execution recording
- 📝 **Traces** - Detailed execution traces
- 📋 **Error Context** - Complete error information

Location: `test-results/` directory

Performance Metrics

Metric	Value
Total Execution Time	1.3 minutes
Average Test Duration	3.2 seconds
Slowest Test	9.0 seconds (Failed tests)
Fastest Test	2.2 seconds
Browser Startup Time	~2 seconds
Parallel Workers	4

Recommendations

Immediate Actions Required

1. ✋ **Fix BUG #3** - Add error handling for negative numbers
2. ✋ **Fix BUG #4** - Add validation for very long numbers
3. ✋ **Fix BUG #2** - Swap Terms and Privacy link hrefs

Short-term Improvements

1. 📝 Add user guidance on acceptable input range
2. 🔧 Fix copyright year JavaScript
3. 🔗 Add target="_blank" to external links
4. ✅ Add input range validation (0-170)

Long-term Enhancements

1. 📊 Implement BigInt for large number support
2. 📱 Add mobile responsive testing
3. ♿ Add accessibility testing
4. 🌐 Add cross-browser testing (Firefox, Safari)

Artifacts Generated

Reports

- ✅ HTML Report: `playwright-report/index.html`
- ✅ JSON Report: `test-results/test-results.json`
- ✅ Bug Report: `docs/BUG-REPORT.md`
- ✅ Test Cases: `docs/TEST-CASES.md`

Evidence

- ✅ Screenshots: 34+ images in test-results/
- ✅ Videos: 3 failure videos in test-results/
- ✅ Traces: Available for failed tests

Conclusion

The automated testing successfully:

- ✅ Validated core functionality (100% pass rate)
- ✅ Identified 8 bugs across different severity levels
- ✅ Achieved 91.2% overall pass rate
- ✅ Completed all 3 special required tests
- ✅ Provided comprehensive documentation
- ✅ Generated detailed evidence and reports

Next Steps:

1. Review and prioritize bug fixes
2. Re-run tests after bug fixes
3. Add additional test coverage for mobile
4. Integrate into CI/CD pipeline

Framework: Playwright + TypeScript + POM **Maintained By:** QA Automation Team **Last Updated:** September 24, 2024